

Evaluating Supply Chain Flexibility of 3PLs

A Linear Algebra Approach

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Introduction

- Objective: Evaluate Third-Party Logistics (3PL) providers using Linear Algebra tools.
- Motivation: Identify flexible and stable logistics partners under varying conditions.
- Approach:
 - Matrices for performance representation
 - Normalization (Min–Max, Z-score)
 - Weighted Scoring
 - PCA + SVD for deeper insights
 - Scenario Analysis (Normal, Peak, Disruption)

Problem Statement

- Companies need structured, mathematical ways to compare logistics providers.
- Evaluate 3PLs across 6 criteria:
 - Cost, Time, Flexibility, Reliability, Capacity, Coverage
- Determine:
 - Best-performing 3PL
 - Stability across Normal, Peak, Disruption
 - Hidden patterns via PCA and SVD

Research Background

- Based on: Choy et al. (2008) – Hybrid Knowledge-Based Logistics Measurement.
- Paper emphasizes flexibility, responsiveness, and reliability.
- Our implementation focuses on:
 - Matrix-based evaluation
 - Scenario analysis
 - PCA/SVD interpretation

Dataset

Code Pipeline

- Dataset: 5 providers $\times$ 6 criteria $\times$ 3 scenarios.
- Python modules created:
 - `data_generator.py` — generates dataset.
 - `normalizer.py` — min-max
z-score normalization.
 - `scorer.py` — weighted scoring.
 - `analyzer.py` — PCA
SVD analysis.
 - `main.py` — complete pipeline runner.
- Outputs:
 - CSV files, PCA plots, scenario comparisons.

Matrix Representation

Normalization

- Each scenario = matrix X (Providers $\times$ Criteria)
- Normalization:
 - **Min-Max**: scales values to $[0,1]$
 - **Z-Score**: centers for PCA/SVD

```
# Min-max normalization
```

```
X_min = X.min(axis=0)
```

```
X_max = X.max(axis=0)
```

```
X_norm = (X - X_min) / (X_max - X_min)
```

```
# Z-score normalization
```

```
from scipy.stats import zscore
```

```
X_z = zscore(X, axis=0)
```

Weighted Scoring (scorer.py)

- Weight vector $\mathbf{w}$ captures importance. Example: [0.25, 0.20, 0.20, 0.15, 0.10, 0.10]
- Final score: $\mathbf{S} = \mathbf{X}_{\text{norm}} \times \mathbf{w}$

```
# Example weight vector (6 criteria)
```

```
w = np.array([0.25,0.20,0.20,0.15,0.10,0.10])
```

```
# Compute scores
```

```
S = X_minmax.dot(w)
```

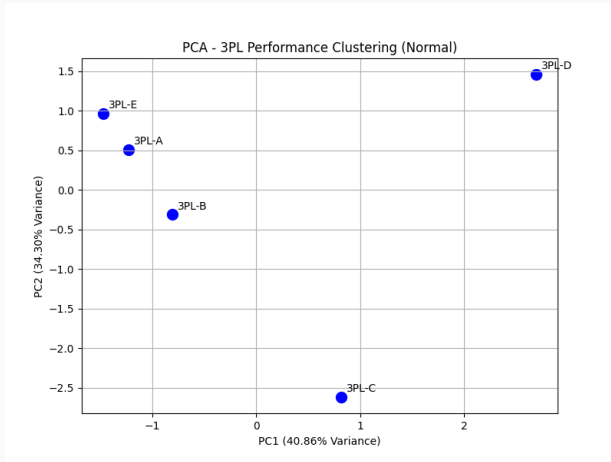
```
# Save results
```

```
df = pd.DataFrame({'Provider': providers, 'Score': S})
```

```
df.to_csv("3PL_scores.csv", index=False)
```

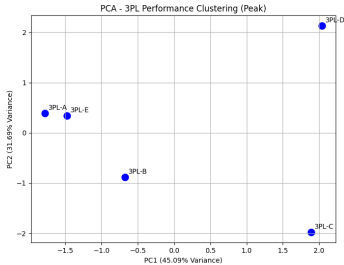
- AHP (eigenvector method) available for deriving weights.

PCA: Normal Scenario

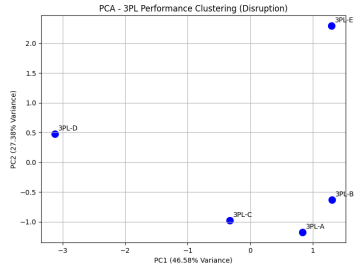


PCA clustering shows similarity and performance grouping.

PCA: Peak Disruption Scenarios



*Peak Scenario



*Disruption Scenario

SVD for Pattern Extraction

- SVD decomposes matrix: $X_c = U\Sigma V^T$
- Σ : reveals strength of hidden performance patterns
- Applications:
 - Robust ranking
 - Noise reduction
 - Understanding criteria influence

```
Xc = X - X.mean(axis=0)
U, S, Vt = np.linalg.svd(Xc, full_matrices=False)
print("Singular Values:", S)
```

Conclusion

References

- Linear Algebra provides a complete evaluation framework.
- Normalization + Scoring + PCA + SVD = powerful insights.
- Scenario analysis reveals true flexibility of 3PLs.
- Approach supports real-world logistics decision-making.

References

- Choy et al., Expert Systems with Applications, 2008.
- Jolliffe, Principal Component Analysis, 2002.
- Golub
Van Loan, Matrix Computations.
- Saaty, Analytic Hierarchy Process.